

Candidate Task Pool

PyTorch Repositories

- model/
- train_or_serve/
- data/
- utils/
- checkpoints/

Candidate Training Tasks

many model families × training algorithms

SFT

DPO

PPO

Candidate Serving Tasks

many serving / scheduling configurations

Core

Ragged

Chunked

Benchmark Construction

Curate / Filter

- ✓ Diversity across model families
- ✓ Bounded configs (size < 7B)
- ✓ Self-contained repo
- ✓ Runnable training / serving pipeline
- ✓ Checkpoint available

retained tasks

Benchmark output

Retained
63 tasks of 18 model families
45 training
18 serving

Per-Task Migration (Prompted Agent)

Min-constraint Prompt (task description+repository) → Coding Agent (with scaffold)

Source



PyTorch Repository

Migrate

(conversion + checkpoint loader)

Target



JAX/Flax Repository

Deliverables (per task)

- ✓ JAX/Flax codebase (torch-free)
- ✓ Checkpoint conversion (from PyTorch weights)
- ✓ Training or serving script that runs end-to-end

Execution-Grounded Evaluation



Isolated Verifier

- torch-free environment
- no network access

Equivalence Verification



① Spec (static)

- model weight
- assertions (e.g., shapes, dtypes)



② Numeric (per-step)

- parameters
- loss
- gradients



③ Behavioral (end-to-end)

- training dynamics (SFT / DPO / PPO)
- decode behavior (paged-attention serving)



Continuous, threshold-free reward